

Exponential Form and Logarithmic Form

Rewriting an exponential statement as a logarithm, rewriting a logarithm as a power, evaluating a logarithm, and solving by switching form

The model: one fact, two ways of writing it

In words	Exponential form	Logarithmic form
2 to the 3rd power is 8	$2^3 = 8$	$\log_2 8 = 3$
10 to the 2nd power is 100	$10^2 = 100$	$\log_{10} 100 = 2$
5 to the 0th power is 1	$5^0 = 1$	$\log_5 1 = 0$
<i>base to the answer is input</i>	$b^y = x$	$\log_b x = y$

Directions

- Read the bottom row of the table both ways: **a logarithm is an exponent**. $\log_b x$ answers the question “what power of b gives x ?”
- The base stays the base when you switch forms. Only the exponent and the input change places.
- Show the switched form as part of your work, then write the number asked for on the blank.

1. The doubling table gives $2^5 = 32$. Rewrite that fact in logarithmic form, and give the value of $\log_2 32$.

Answer: _____

2. A logarithm statement says $\log_5 x = 3$. Rewrite it in exponential form, and give the value of x .

Answer: _____

3. Evaluate $\log_4 64$ by asking which power of 4 gives 64.

Answer: _____

4. Negative exponents work the same way.

(a) Give the value of 7^{-2} as a fraction: _____

(b) Using part (a), give the value of $\log_7\left(\frac{1}{49}\right)$: _____

5. A logarithm statement says $\log_3 x = -2$. Rewrite it in exponential form, and give the value of x as a fraction.

Answer: _____

6. Solve for x by writing both sides as powers of the same base: $2^x = \frac{1}{32}$

Answer: _____

7. Evaluate $\log_9 3$. (What power of 9 gives 3?)

Answer: _____

8. A bacteria culture doubles every hour. Starting from a single cell, the count after t hours is $N = 2^t$. A technician counts 4096 cells. Write the logarithmic form of that statement, and find how long the culture has been growing.

Answer: _____ hours

9. A logarithm statement says $\log_b 216 = 3$, where b is the base. Rewrite it in exponential form and find the base b .

Answer: _____

10. Challenge. Consider $\log_2(3x - 1) = 4$.

(a) Rewrite it in exponential form and solve. Give the exact value of x : _____

(b) Give that solution as a decimal, rounded to two places: _____

(c) Explain why no value of x at or below $\frac{1}{3}$ could ever be a solution of this equation.